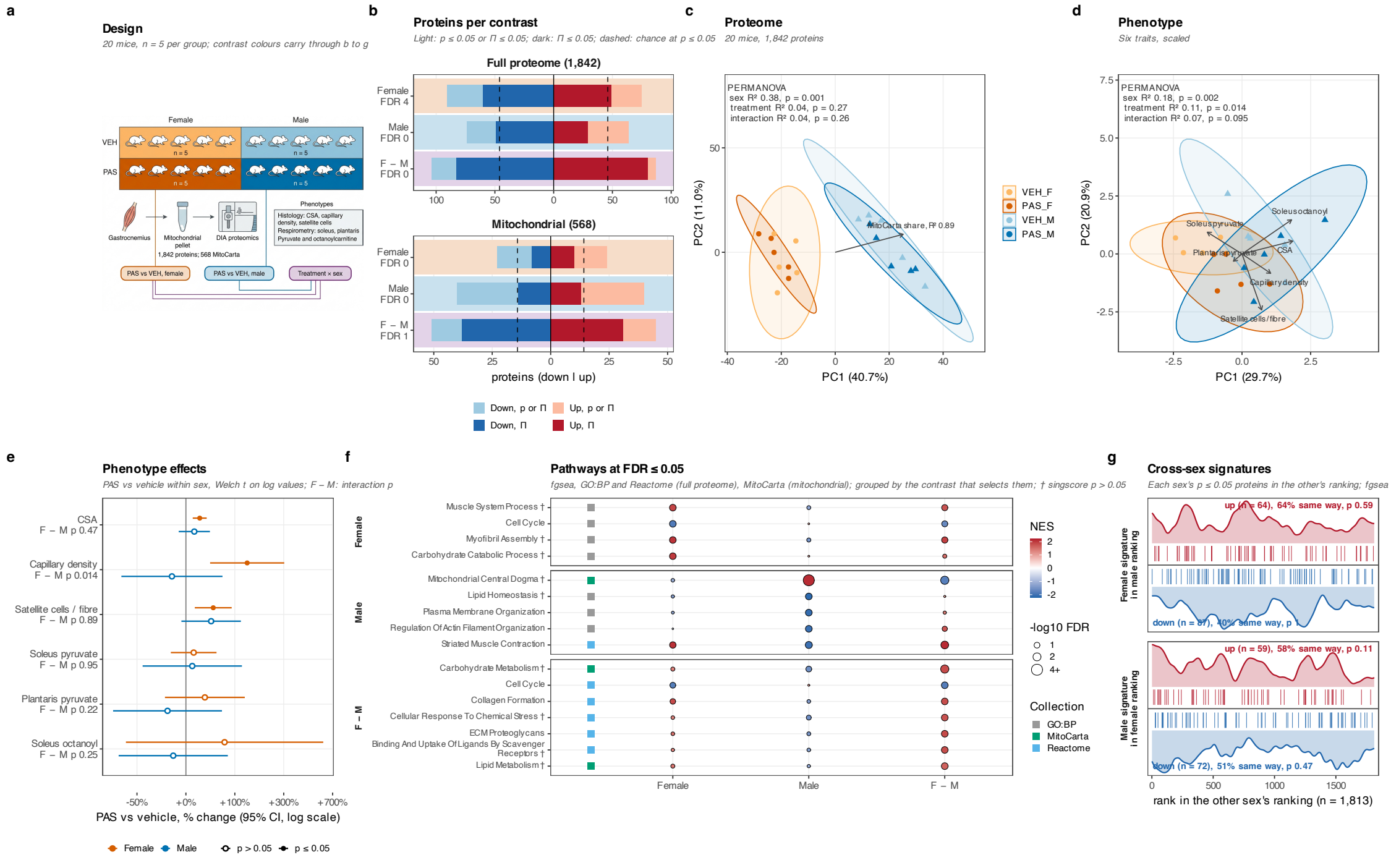

a
Design
20 mice, n = 5 per group; contrast colours carry through b to g
Female
Male
VEH
PAS
n = 5
Gastrocnemius
Mitochondrial pellet
DIA proteomics
1,842 proteins; 568 MitoCarta
Phenotypes
Histology: CSA, capillary density, satellite cells
Respirometry: soleus, plantaris
Pyruvate and octanoylcarnitine
PAS vs VEH, female
PAS vs VEH, male
Treatment × sex
b
Proteins per contrast
Light: p ≤ 0.05 or Π ≤ 0.05; dark: Π ≤ 0.05; dashed: chance at p ≤ 0.05
Full proteome (1,842)
Female FDR 4
Male FDR 0
F − M FDR 0
Mitochondrial (568)
Female FDR 0
Male FDR 0
F − M FDR 1
proteins (down | up)
Down, p or Π
Up, p or Π
Down, Π
Up, Π
c
Proteome
20 mice, 1,842 proteins
PERMANOVA
sex R² 0.38, p = 0.001
treatment R² 0.04, p = 0.27
interaction R² 0.04, p = 0.26
MitoCarta share, R² 0.89
PC1 (40.7%)
PC2 (11.0%)
VEH_F
PAS_F
VEH_M
PAS_M
d
Phenotype
Six traits, scaled
PERMANOVA
sex R² 0.18, p = 0.002
treatment R² 0.11, p = 0.014
interaction R² 0.07, p = 0.095
Soleus octanoyl
Soleus pyruvate
Plantaris pyruvate
CSA
Capillary density
Satellite cells / fibre
PC1 (29.7%)
PC2 (20.9%)
e
Phenotype effects
PAS vs vehicle within sex, Welch t on log values; F − M: interaction p
CSA
F − M p 0.47
Capillary density
F − M p 0.014
Satellite cells / fibre
F − M p 0.89
Soleus pyruvate
F − M p 0.95
Plantaris pyruvate
F − M p 0.22
Soleus octanoyl
F − M p 0.25
-50%
+0%
+100%
+300%
+700%
PAS vs vehicle, % change (95% CI, log scale)
Female
Male
p > 0.05
p ≤ 0.05
f
Pathways at FDR ≤ 0.05
fgsea, GO:BP and Reactome (full proteome), MitoCarta (mitochondrial); grouped by the contrast that selects them; † singscore p > 0.05
Female
Male
F − M
Muscle System Process †
Cell Cycle
Myofibril Assembly †
Carbohydrate Catabolic Process †
Mitochondrial Central Dogma †
Lipid Homeostasis †
Plasma Membrane Organization
Regulation Of Actin Filament Organization
Striated Muscle Contraction
Carbohydrate Metabolism †
Cell Cycle
Collagen Formation
Cellular Response To Chemical Stress †
ECM Proteoglycans
Binding And Uptake Of Ligands By Scavenger Receptors †
Lipid Metabolism †
NES
-log10 FDR
Collection
GO:BP
MitoCarta
Reactome
g
Cross-sex signatures
Each sex's p ≤ 0.05 proteins in the other's ranking; fgsea
Female signature in male ranking
up (n = 64), 64% same way, p 0.59
down (n = 87), 40% same way, p
Male signature in female ranking
up (n = 59), 58% same way, p 0.11
down (n = 72), 51% same way, p 0.47
rank in the other sex's ranking (n = 1,813)